

GDOR-SLAM: Guarded Dynamic Observation Recovery for RGB-D Gaussian SLAM

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Abstract—Dynamic masks protect a persistent Gaussian map, but hard exclusion can also remove background and camera-consistent observations that remain useful for localization. This paper studies a task mismatch in dynamic RGB-D Gaussian SLAM: an observation that is useful for transient camera tracking is not necessarily eligible to modify a static world model. We present GDOR-SLAM, which implements separate tracking-support and persistent-mapping routes for dynamic-region evidence. Temporal depth consistency, residual-flow consistency, and tracking-risk state qualify a conservative subset of excluded observations for current-frame feature extraction and pose estimation, while the original conservative mask continues to govern ORB landmark insertion and Gaussian initialization, optimization, and densification. The method therefore recovers tracking support without granting recovered dynamic-region observations persistent map authority. The completed DYN-19 release contains a 90-run main comparison over ten sequences and three seeds, a 120-run ordered ablation, and a 72-cell multi-seed mapping diagnostic; earlier frozen DYN-15 and DYN-18 campaigns provide complementary validation. Against semantic hard exclusion in DYN-19, GDOR lowers mean ATE from 10.119 cm to 4.930 cm (51.3%); the paired-cell median gain is 0.213 cm, the win/tie/loss count is 19/0/11, and the largest failure-case rescue is Bonn crowd2 at 42.500 cm to 11.080 cm. Against the map-matched no-track-reuse control, the mean decreases from 7.392 cm to 4.930 cm, but the paired-cell median gain is only 0.016 cm with a 16/0/14 record and a 4/0/6 record over sequence means; this aggregate is driven by sitting_halfsphere at 36.775 cm to 4.864 cm. All 30 GDOR cells pass the recovered-support overlap certificate, covering 79.498M audited pixels with no observed map-admission overlap under the declared route invariant. This certificate establishes route behavior, not zero ghost contamination; the ordered ablation does not identify independent component effects or interactions, and protocol-matched external-baseline and independent ghost/completeness claims remain pending.

Index Terms—3D Gaussian splatting, dynamic RGB-D SLAM, visual tracking, dynamic observation recovery, Gaussian mapping.

I. INTRODUCTION

3D Gaussian Splatting (3DGS) provides an explicit differentiable representation for dense scene rendering and has become a practical substrate for online visual world modeling [1]. Gaussian SLAM systems such as Photo-SLAM, SplatAM, GS-SLAM, and Gaussian-SLAM combine camera tracking with dense Gaussian reconstruction [2]–[5]. Their tracking frontends and persistent maps, however, still depend on a predominantly static world. When people or objects occupy a large image region, dynamic observations can corrupt camera estimation and produce persistent Gaussian artifacts.

Semantic and geometric exclusion is therefore a natural defense. Classical dynamic SLAM removes features associated with movable classes [6]–[8], while recent dynamic Gaussian

SLAM systems combine semantics, temporal consistency, depth warping, photometric evidence, or uncertainty to suppress unsafe observations [9]–[12]. This policy is appropriate for static map construction, but it can be too restrictive for tracking. A semantic mask can cover visible background, object boundaries, temporarily stationary surfaces, or pixels whose image motion is dominated by camera motion. Removing all such observations can leave the pose estimator poorly constrained exactly when dynamic occlusion is strongest.

The key issue is not only whether an observation is static, but what authority it should receive. Tracking and mapping have different failure costs and persistence. A locally consistent observation may be useful for estimating the current camera pose while still being too uncertain to initialize or optimize persistent static structure. Treating both decisions with one binary mask conflates *tracking utility* with *map eligibility*.

We propose **GDOR-SLAM**, a dynamic RGB-D Gaussian SLAM system based on Guarded Dynamic Observation Recovery (GDOR). GDOR evaluates temporal depth consistency, residual optical-flow consistency, and the current tracking-risk state to recover a conservative subset of observations inside excluded dynamic regions. Recovered observations may replenish features and support the current camera estimate. They do not modify the conservative mask used for persistent ORB landmarks or Gaussian initialization, update, and densification. The method therefore separates transient tracking privilege from persistent map privilege.

Our contributions are:

- We instantiate an auditable two-channel observation contract for RGB-D/ORB Gaussian SLAM: tracking utility does not automatically grant persistent Gaussian map authority.
- We introduce GDOR, which combines temporal depth, residual-flow, and tracking-risk evidence to selectively recover dynamic-region observations for transient tracking.
- We route the recovered tracking mask and conservative mapping weight independently; the release implements explicit recovered-support overlap, MapPoint-rejection, and RGB-D densification-rejection audits.
- We release a completed DYN-19 package with 90 main runs, 120 ordered-ablation runs, 72 multi-seed mapping cells, and per-run route certificates, together with the earlier frozen DYN-15–18 evidence.

The supported claim is deliberately bounded. GDOR improves the DYN-19 aggregate relative to the same-source Semantic and map-matched routes, but the paired results are non-uniform and concentrated in severe failure regimes. The multi-seed mapping package supplies route and rendering di-

GDOR-SLAM: guarded tracking support without persistent dynamic writes

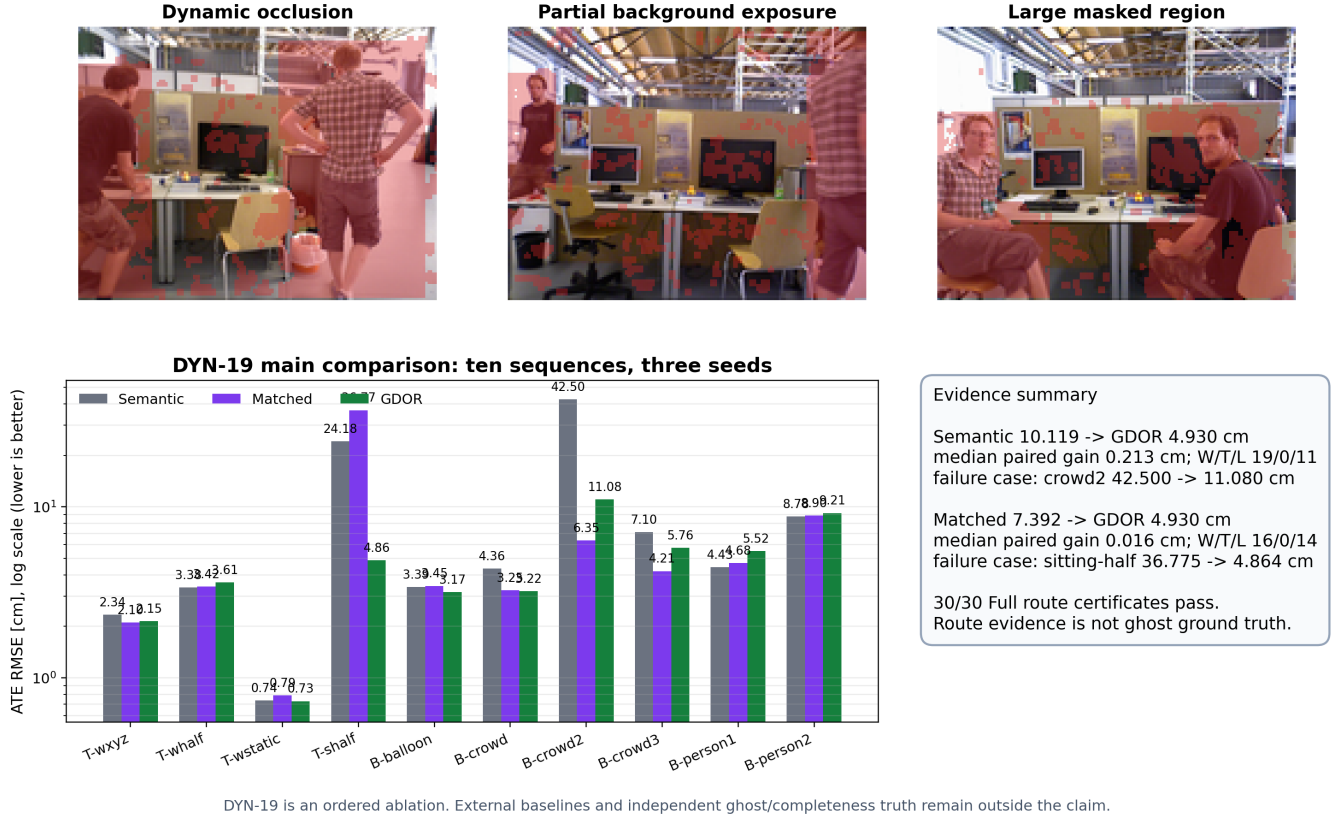


Fig. 1. GDOR-SLAM overview and DYN-19 main evidence. Real TUM dynamic frames show large semantic exclusion regions. The ten-sequence, three-seed comparison exposes both failure-regime rescue and non-uniform cell outcomes against Semantic and the map-matched control. Published external-method values remain context rather than protocol-matched rows.

agnostics, while its occluded-background fields remain proxies rather than independent ghost or completeness truth. The paper therefore argues for a concrete RGB-D/ORB recovery and routing mechanism rather than conceptual priority, independent component causality, universal improvement, or external SOTA superiority.

II. RELATED WORK

A. Visual SLAM in Dynamic Environments

Classical feature and dense SLAM systems rely on robust estimation but still assume that most observations belong to a static scene [13]–[16]. Semantic dynamic SLAM makes this assumption explicit by removing features or pixels associated with potentially moving classes. DynaSLAM combines learned instance masks with multi-view geometry [6], while DS-SLAM and Dynamic-SLAM use semantic predictions to reject dynamic correspondences [7], [8]. Residual-based reconstruction instead detects observations that disagree with the estimated static model, as in ReFusion [17].

Motion-aware systems retain more information about the moving scene. VDO-SLAM estimates camera and object motion from dynamic observations [18], and optical-flow tracking provides a standard source of short temporal correspondences [19]. The central difficulty is identifiability:

image motion alone mixes camera motion, object motion, and correspondence error. Our method uses camera-motion inconsistency as a guarded cue, but the main distinction is role assignment: recovered dynamic-region observations may support short-term tracking while remaining ineligible for persistent static-map landmarks.

B. Dense Neural and Gaussian SLAM

Dense neural SLAM has represented scene geometry with implicit fields or feature grids [20]–[22], while 3DGS provides an explicit representation with differentiable rasterization [1]. Photo-SLAM couples ORB-SLAM3 with a Gaussian mapper [2]; SplaTAM jointly tracks and maps RGB-D observations with Gaussians [3]; GS-SLAM and Gaussian-SLAM develop online Gaussian densification and optimization strategies [4], [5]. These methods establish dense Gaussian mapping in static scenes but do not by themselves resolve whether a measurement should influence camera tracking, map optimization, both, or neither in a dynamic scene.

C. Dynamic Gaussian SLAM

The closest static-output Gaussian SLAM systems suppress dynamic evidence using combinations of semantics, temporal consistency, and geometry. DG-SLAM fuses semantic and

multi-frame depth-warp masks and refines a coarse pose against the Gaussian map [9]. DGS-SLAM refines segmentation through photometric consistency and filters dynamic content during Gaussian insertion and keyframe selection [10]. DyGS-SLAM improves dynamic-scene localization and Gaussian reconstruction in real time [23], and MPDG-SLAM introduces motion-probability reasoning for dynamic 3DGS-SLAM [24]. DyPho-SLAM uses refined masks and adaptive feature extraction for real-time dynamic Gaussian SLAM [11]. DAGS-SLAM maintains a spatiotemporal motion probability for each Gaussian and schedules semantic inference from uncertainty [12]. Concurrent DL-SLAM explicitly uses transiently static objects for pose estimation while suppressing their persistent contribution through dual-level probabilities [25]. This establishes tracking–mapping role separation as a shared recent direction rather than a conceptual priority claim of GDOR.

Dynamic-output systems address a different objective. D²GSLAM combines static 3D Gaussians with dynamic 4D Gaussians and uses motion information for progressive pose refinement [26]. Our output remains a static Gaussian map. Rather than reconstructing object trajectories, estimating object-level probabilities, or injecting object-motion priors, GDOR instantiates a narrower RGB-D/ORB mechanism: binary local recovery from temporal depth and residual-flow checks, separate `staticMask/staticWeight` routes, hard persistent-weight gating for MapPoints and Gaussians, and an explicit recovered-support overlap audit. The contribution is this auditable implementation and its matched no-track-reuse control, not the first statement that tracking and mapping can use different dynamic evidence.

III. METHOD

A. Observation Routing Contract

GDOR-SLAM extends the RGB-D path of Photo-SLAM [2], with an ORB-SLAM3 frontend [14] and a 3DGS mapping backend [1]. All masks below are binary and defined per pixel $\mathbf{x}$. Let $M_t^{\text{sem}} = 1$ denote semantic exclusion and $M_t^{\text{flow}} = 1$ denote residual-motion exclusion. Their union and its morphological closing are

$$D_t^{\text{raw}} = M_t^{\text{sem}} \vee M_t^{\text{flow}}, \quad \bar{D}_t = \text{Close}(D_t^{\text{raw}}). \quad (1)$$

The claim-bearing configuration routes two different signals: a binary support mask S_t^{track} to ORB extraction and current-frame tracking, and a float weight W_t^{map} to persistent ORB/Gaussian mapping. The conservative mapping route is fixed as

$$W_t^{\text{map}}(\mathbf{x}) = 1 - \bar{D}_t(\mathbf{x}). \quad (2)$$

Thus, a tracking decision can recover support without clearing the persistent mapping exclusion.

B. Residual-Flow and Temporal-Depth Evidence

A semantic prior marks classes that may move. Forward–backward Lucas–Kanade tracks [19] are sampled on an image grid; tracks with more than 1.5 pixels of round-trip error are discarded. A partial affine model is fitted by RANSAC

on semantic-static support, while a fundamental-matrix check requires at least six model correspondences and at least 60% epipolar inliers. If either model is invalid, flow-based recovery fails closed. Otherwise, residuals unexplained by the affine model form M_t^{flow} , and the full-frame residual mask plus a safety neighborhood forms the recovery veto G_t .

Let z_t , z_t^r , and b_{t-1} be observed, rendered, and historical background depths. For valid inputs, the temporal prior is

$$b_t(\mathbf{x}) = \frac{w_h I_h b_{t-1}(\mathbf{x}) + w_r I_r z_t^r(\mathbf{x})}{w_h I_h + w_r I_r}, \quad (3)$$

where I_h and I_r indicate valid historical and rendered depth. Depth consistency is accepted when rendered support is valid and at least K valid observations in a local neighborhood $\mathcal{N}(\mathbf{x})$ satisfy

$$C_t^{\text{depth}}(\mathbf{x}) = \left[\sum_{\mathbf{y} \in \mathcal{N}(\mathbf{x})} \mathbf{1}(|z_t(\mathbf{y}) - b_t(\mathbf{x})| < \tau_d) \geq K \right]. \quad (4)$$

C. Guarded Support Qualification

A raw-excluded pixel is eligible for tracking recovery only when every required signal is valid:

$$R_t^{\text{elig}}(\mathbf{x}) = D_t^{\text{raw}}(\mathbf{x}) \wedge C_t^{\text{depth}}(\mathbf{x}) \wedge \neg G_t(\mathbf{x}). \quad (5)$$

After the same morphological refinement used by the frontend, the tracking support is

$$S_t^{\text{track}}(\mathbf{x}) = \neg \text{Close}(D_t^{\text{raw}} \wedge \neg R_t^{\text{elig}})(\mathbf{x}), \quad (6)$$

and the support actually recovered after refinement is $R_t = D_t^{\text{raw}} \wedge S_t^{\text{track}}$. Invalid depth, insufficient rendered support, or an invalid motion model sets $R_t^{\text{elig}} = 0$ and returns the conservative raw-mask behavior.

D. Risk-Conditioned Tracking Recovery

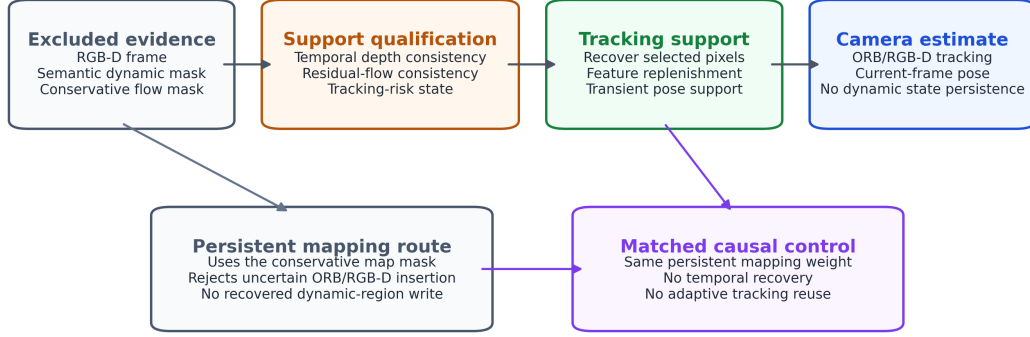
Recovery is useful only when the tracker needs additional support. GDOR therefore conditions feature replenishment on the previous optimized inlier count. During a tracking-risk window, the FAST threshold is temporarily relaxed and qualified pixels can participate in feature extraction and current-frame pose estimation. The relaxation persists for a short fixed hold interval to avoid frame-to-frame switching. Outside these windows, the system remains close to semantic hard exclusion.

The default Guarded configuration uses residual-flow protection, temporal-depth recovery, and risk-conditioned feature replenishment. The Strict configuration adds an RGB-D camera-guided flow model and tighter model-quality gates. Strict is retained as a stronger-intervention control, not as the proposed default, because stronger recovery is not uniformly more stable in the broader TUM/Bonn protocol.

E. Tracking–Mapping Decoupling and Matched Control

Recovered observations are transient. They may support current-frame ORB extraction and pose optimization, but the mapping route continues to use Eq. (2) for Gaussian loss weighting, RGB-D densification, and MapPoint-to-Gaussian

GDOR-SLAM: guarded recovery with separate tracking support and mapping weight



Core contract: tracking support does not automatically grant persistent Gaussian map admission.

Fig. 2. GDOR-SLAM separates support qualification, transient tracking support, and persistent mapping weight. The matched no-track-reuse control uses the same persistent weight as GDOR but disables temporal recovery and adaptive tracking reuse.

admission. The route invariant is frame-local and admission-local: a frame that contains recovered tracking support does not receive a persistent-map admission privilege. Every ORB MapPoint, Gaussian initialization, RGB-D densification candidate, and Gaussian loss term is still admitted or weighted only through W_t^{map} ; recovered support cannot set that weight to one, bypass keyframe static-weight checks, or waive the conservative mapping boundary. The implemented routing cases are summarized compactly as

configuration	S_t^{track}	W_t^{map}	
Semantic	$1 - \text{Close}(M_t^{\text{sem}})$	1 (default)	(7)
Matched	$1 - \bar{D}_t$	$1 - \bar{D}_t$	
GDOR/Strict	Eq. (6)	$1 - \bar{D}_t$	

The Semantic row has no supplied static weight, so the mapper reports its default unit support; this does not mean that the semantic tracking mask contains no exclusions.

The release audits the exact overlap

$$L_t = R_t \wedge \mathbf{1}[W_t^{\text{map}} \geq 0.5]. \quad (8)$$

For the claim-bearing routing, the intended invariant is $|L_t| = 0$. The counter is diagnostic and does not modify either route. Persistent ORB MapPoints are additionally checked against keyframe static weights, while Gaussian training and RGB-D densification consume the same bound weight.

To isolate tracking-side recovery from the raw-mask route, the map-matched no-track-reuse control uses the same $W_t^{\text{map}} = 1 - \bar{D}_t$ as GDOR, uses $1 - \bar{D}_t$ directly for tracking, and disables temporal recovery and adaptive feature replenishment. The GDOR–Matched comparison therefore tests qualified tracking reuse on top of the same persistent mapping weight; it is not a comparison between two Gaussian-admission masks.

F. Implementation Details

The frozen Guarded configuration applies semantic inference on every frame after a 50-frame system warmup and starts

temporal background refinement after five frames. Lucas–Kanade tracking uses a 21×21 window, three pyramid levels, and an 8-pixel sampling grid. The dominant-flow residual threshold is 1.5 pixels. Temporal depth combines rendered and historical background estimates with weights 0.6 and 0.4; accepted static observations update the stored background with weight 0.2. The local consistency threshold is 0.20 m. The local test uses a radius of two pixels, requires nine consistent neighbors, a minimum valid fraction of 0.02, and rendered opacity of at least 0.5. The prediction is disabled when the inter-frame motion exceeds 0.5 m translation or 0.7854 rad rotation.

Feature replenishment is activated when the previous optimized inlier count falls below 150. The FAST threshold is relaxed by a factor of 0.9 and the state is held for five frames. The flow guard uses a one-pixel conservative safety radius. All positive experiments keep object-motion pose priors disabled, use hard mapping masks, and prohibit soft dynamic Gaussian admission. These values were frozen before DYN-18 and were not adjusted using its outcomes. The added per-frame work is sparse LK/model fitting plus linear-time mask and depth tests; GDOR introduces no object state, pose prior, or additional persistent map variable.

IV. EXPERIMENTS

A. Protocols and Metrics

We evaluate GDOR-SLAM on dynamic RGB-D sequences from TUM RGB-D [27] and Bonn RGB-D Dynamic [17]. The completed DYN-19 release is the primary evidence package. Its main phase contains Semantic, MapMatched, and Full on ten claim-bearing sequences and seeds 0–2, for 90 runs; the denominator includes TUM sitting_halfsphere. The mechanism phase adds T+M, T+F+M, T+F+R+M, and Full-NoM on the same sequence/seed matrix, for 120 runs. Here T, F, R, A, and M denote temporal-depth recovery, the residual-flow recovery veto, tracking-risk gating, adaptive FAST, and

conservative persistent mapping weight. The seven configurations form a predeclared *ordered ablation*, not a factorial or fractional-factorial design. The mapping phase evaluates Semantic, MapMatched, and Full over four scenes, three seeds, and online/GT-aligned render poses, for 72 cells.

Earlier frozen campaigns remain separate corroborating evidence. **DYN-15** contains Semantic, GDOR, and Strict with three seeds on TUM walking_xyz and all six Bonn dynamic sequences, for 63 runs. **DYN-18** uses the unchanged configurations on the four TUM sequences named in the DyPho-SLAM tracking table, for 36 cells after one exact Semantic sitting seed-0 repair. **DYN-17** is the three-sequence predecessor of the DYN-19 map-matched control. **DYN-16** is a seed-0 common-view mapping diagnostic on TUM walking_xyz and Bonn crowd and is retained only as a limited preservation diagnostic.

All tracking runs process the full association file and retain invalid intervals in the all-input-frame trajectory. We report SE(3)-aligned ATE RMSE, one-frame translational RPE RMSE, failure rate, and end-to-end wall time. Mapping uses common non-keyframe manifests and reports full/static PSNR and SSIM; LPIPS is unavailable in the DYN-19 release and is not imputed. Occluded-background completeness and ghost-risk values are reported only as proxy diagnostics.

The benchmark harness records the source snapshot, executable and library hashes, detector-engine hash, configuration hash, association and ground-truth hashes, sequence, seed, command line, and output identity. DYN-19 binds every result to source commit 14c6b2c, frozen plan hashes, per-frame logs, run summaries, and recovered-support overlap certificates. The public release contains trajectories, masks, maps, renders, per-frame logs, certificates, plans, summaries, and SHA-256 manifests.

B. Tracking Accuracy and Frozen Generalization

In DYN-15, GDOR reduces the All7 mean ATE from 11.331 cm to 3.798 cm, shifts the sequence-median ATE from 4.278 cm to 3.261 cm, and wins every declared sequence mean relative to Semantic (7/0/0 wins/ties/losses). The largest rescue is Bonn crowd2 at 52.184 cm to 3.226 cm. DYN-18 yields a 9/0/3 paired-cell record and a 0.209 cm median paired gain, but its mean change from 9.746 cm to 3.415 cm is driven mainly by sitting_halfsphere at 32.374 cm to 7.354 cm. The two-sided exact sign test is not significant ($p = 0.146$), so DYN-18 is interpreted as conditional failure-regime rescue plus near-identity behavior elsewhere.

C. DYN-19 Main Extension and Ordered Ablation

Table II broadens the matched control to all ten claim-bearing sequences. Relative to Semantic, Full lowers the 30-cell mean from 10.119 cm to 4.930 cm, with a 0.213 cm paired-cell median gain and a 19/0/11 cell record; the largest failure-case rescue is Bonn crowd2 at 42.500 cm to 11.080 cm. Relative to MapMatched, Full lowers the mean from 7.392 cm to 4.930 cm, but the paired median is only 0.016 cm, the cell record is 16/0/14, and Full wins only four of ten sequence means. That aggregate is dominated by sitting_halfsphere, where MapMatched reaches 36.775 cm and Full reaches 4.864

cm. The supported interpretation is therefore severe-failure rescue rather than uniform improvement.

Panel B is deliberately non-monotonic. T+M has the lowest overall mean, the flow-veto row is worse in mean despite a 6/0/4 sequence-mean record against T+M, and adding adaptive FAST changes the mean only from 4.907 cm to 4.930 cm with a 15/0/15 paired-cell split. These adjacent bundle contrasts motivate the modules operationally but do not estimate independent causal effects. Full-NoM obtains lower tracking ATE than Full, yet all 30 of its route certificates fail by construction because recovered support is admitted to persistent mapping; tracking accuracy alone is therefore insufficient evidence for the routing safety claim.

D. Safety, Route Certificate, and Mapping Diagnostic

GDOR lowers the DYN-15 All7 failure rate but increases wall time from 32.46 s/run to 49.84 s/run. Map support is the mean fraction of pixels satisfying $W_t^{\text{map}} \geq 0.5$, not a quality score. The archived DYN-15–18 rejection counters show active static-weight gates, but only DYN-19 contains the direct overlap certificate from Eq. (8).

Across the 12 online mapping cells per method, Full raises static PSNR from 18.866 dB for Semantic to 19.763 dB; the mean gain is 0.896 dB, the paired median gain is 0.544 dB, and the record is 9/0/3; the lowest-Semantic failure cell is sitting_halfsphere seed 1 at 14.275 dB versus 16.551 dB for Full. Against MapMatched, the corresponding mean is 18.990 dB to 19.763 dB, with a 0.515 dB paired median gain and an 11/0/1 record; its lowest cell is the same sitting sequence/seed at 14.009 dB versus 16.551 dB. The GT-aligned rows are still end-to-end diagnostics because online poses constructed each map. The proxy columns are non-monotonic in online mode and do not support a true ghost-contamination or complete-background claim.

All 30 Full cells pass the recovered-support certificate, covering 3,751 recovered-support frames and 79,498,476 audited pixels with zero observed overlap against map-admissible support. The T+M, T+F+M, and T+F+R+M rows also have 30 passing certificates each. The 30 expected Full-NoM failures contain 90,790,462 overlapping pixels, providing a negative counterfactual that distinguishes the route invariant from a vacuous rejection counter.

The DYN-16 figure is retained as historical evidence only. It uses two scenes, one seed, fixed static masks, and no independent ghost/completeness truth, so it remains a limited preservation diagnostic rather than a zero-ghost, zero-contamination, or broad mapping-superiority result.

E. External Positioning and Limitations

Published DGS-SLAM, DG-SLAM, DyPho-SLAM, and DynaSLAM values remain related-work context rather than protocol-matched DYN-19 rows. The audited DG-SLAM runner uses ground-truth poses inside its shipped Gaussian mapping path, while the audited DynaSLAM source lacks an accepted ten-sequence TUM/Bonn execution path. Both therefore remain outside the main table until online-pose, association, mask-provenance, artifact, and no-failure eligibility

TABLE I

EARLIER SAME-SOURCE TRACKING EVIDENCE. VALUES ARE THREE-SEED ATE RMSE IN CENTIMETERS; LOWER IS BETTER. EVERY REPORTED MEAN IMPROVEMENT IS ACCOMPANIED BY ITS FAILURE CASE, MEDIAN, AND WIN/TIE/LOSS COUNT.

Evidence	Scope	Cells represented	Semantic	GDOR	Strict	GDOR reduction	Failure/median context	W/T/L
DYN-15	Bonn6 mean	54	12.837	4.079	8.526	68.2%	crowd2 52.184→3.226; med. 4.988→3.577	6/0/0
DYN-15	All7 mean	63	11.331	3.798	7.611	66.5%	crowd2 52.184→3.226; med. 4.278→3.261	7/0/0
DYN-18	TUM4 mean	36	9.746	3.415	3.345	65.0%	sitting 32.374→7.354; med. paired gain 0.209	9/0/3

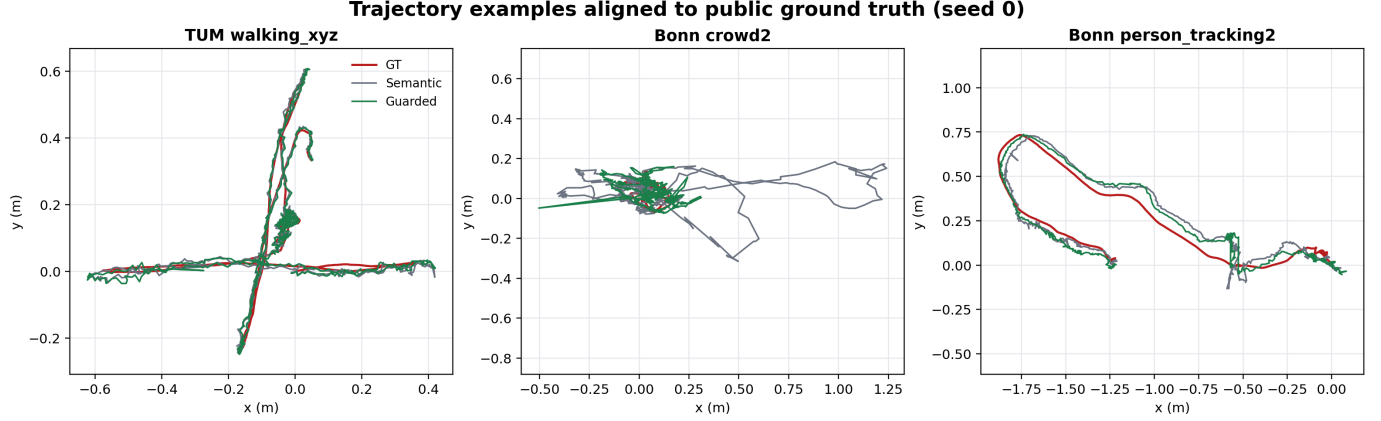


Fig. 3. Seed-0 trajectory examples aligned to public ground truth from the DYN-15 experiment family. The plots are qualitative diagnostics; Table I uses all input frames and three-seed means.

TABLE II

COMPLETED DYN-19 TRACKING RESULTS. PANEL A REPORTS THE TWO CLAIM-BEARING FULL COMPARISONS OVER 30 PAIRED CELLS; SEQUENCE W/T/L IS COMPUTED AFTER AVERAGING THE THREE SEEDS. PANEL B IS AN ORDERED ABLATION. ITS ADJACENT CONTRASTS DO NOT IDENTIFY INDEPENDENT COMPONENT EFFECTS OR INTERACTIONS. POSITIVE ADJACENT MEDIAN MEAN THE CURRENT ROW HAS LOWER PAIRED-CELL ATE THAN THE PRECEDING ROW.

Panel A comparison	Cells	Baseline	Full	Reduction	Failure case	Paired median gain	Cell W/T/L	Sequence W/T/L
Semantic → Full	30	10.119	4.930	51.3%	crowd2 42.500→11.080	0.213 cm	19/0/11	7/0/3
MapMatched → Full	30	7.392	4.930	33.3%	sitting 36.775→4.864	0.016 cm	16/0/14	4/0/6

Panel B configuration	T	F	R	A	M	Mean ATE	Median ATE	Adjacent median; W/T/L	Route certificate
Semantic	0	0	0	0	0	10.119	4.348	–	30 vacuous
MapMatched	0	0	0	0	1	7.392	3.824	+0.211; 18/0/12	30 vacuous
T+M	1	0	0	0	1	3.859	3.489	+0.023; 17/0/13	30 pass
T+F+M	1	1	0	0	1	6.377	3.748	–0.020; 15/0/15	30 pass
T+F+R+M	1	1	1	0	1	4.907	3.721	–0.024; 14/0/16	30 pass
Full: T+F+R+A+M	1	1	1	1	1	4.930	3.640	–0.006; 15/0/15	30 pass
Full-NoM	1	1	1	1	0	4.060	3.627	+0.025; 18/0/12	30 fail (expected)

gates pass. The DyPho-SLAM paper reports 1.6, 2.6, 0.6, and 1.6 cm on the four corresponding TUM sequences; GDOR is not below those external values in DYN-18, and protocol differences prevent a direct ranking.

DYN-19 closes the earlier multi-seed and common-manifest mapping gap, but not the independent-truth gap. Its occluded-background proxy is constructed from the same RGB-D sequences and frozen semantic masks. Blind manual labels or external revealed-background truth with agreement/adjudication and provenance certificates are still required before claiming measured ghost contamination or complete-background recovery.

V. DISCUSSION

A. Why Privilege Separation Matters

Dynamic multimedia capture exposes a persistence asymmetry. A tracking error affects the current pose and may propagate to later states, whereas an unsafe Gaussian write can remain visible from many future viewpoints. Hard exclusion minimizes the second risk but may increase the first when a dynamic mask removes most of the image support. GDOR does not attempt to classify every recovered pixel as truly static. Instead, it asks a narrower question: whether the pixel is sufficiently consistent with the current camera-induced motion and temporal background to support transient localization. This narrower decision is why recovery can be useful without

Failure interval, recovered tracking support, and matched control

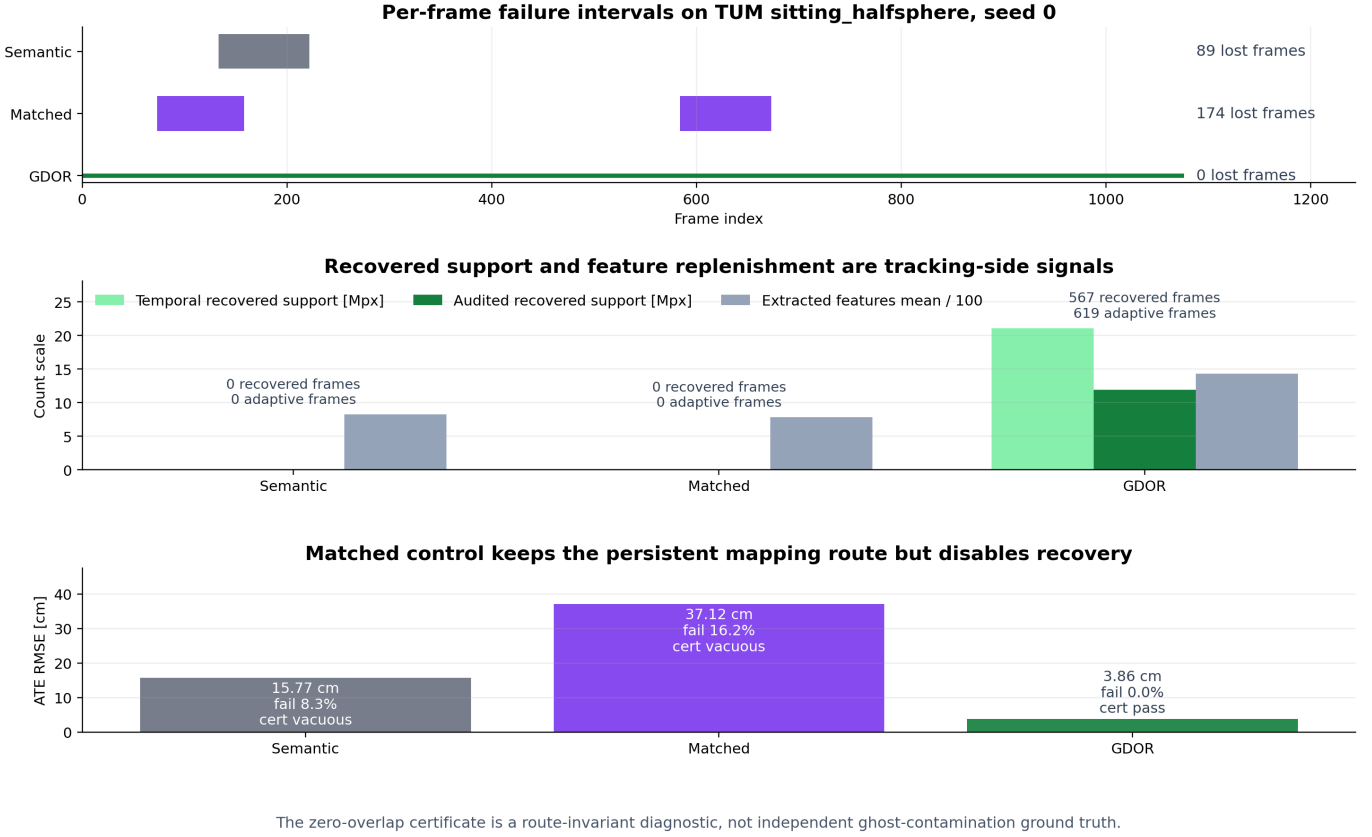


Fig. 4. Representative DYN-19 route diagnostic on TUM sitting_halfsphere, seed 0. Top: failure intervals from per-frame tracking logs. Middle: recovered support and adaptive feature activity are observed only in Full; these are tracking-side signals. Bottom: MapMatched shares the persistent mapping weight but disables qualified tracking recovery. The Full certificate records 567 recovered-support frames, 11.918M audited pixels, and no observed overlap with map-admissible support in this episode; this is a route diagnostic, not independent ghost-contamination truth.

changing the static-map contract.

The completed DYN-19 matched control sharpens this interpretation while exposing its regime dependence. MapMatched shares GDOR’s raw semantic-plus-flow support and persistent mapping weight but disables temporal recovery and adaptive replenishment. Full is substantially better on the sitting_halfsphere failure regime, yet it wins only four of ten sequence means and 16 of 30 paired cells. The evidence therefore supports conditional tracking-side rescue on top of a fixed mapping route, not uniform superiority.

B. Regime Dependence and Failure Boundaries

The results do not imply that every masked region should be recovered. On DYN-18 walking_static and walking_halfsphere, the Semantic parent is already stable and GDOR remains close to it. The larger gain occurs on sitting_halfsphere, where semantic exclusion removes a large portion of the image and the parent experiences repeated invalid intervals. This pattern is consistent with conditional rescue: recovery is most valuable when exclusion creates an observability deficit.

The ordered ablation exposes another boundary. T+M has the lowest overall DYN-19 tracking mean, while subsequent

flow, risk, and adaptive stages are non-monotonic across cells and sequences. Because the schedule is not factorial, these adjacent contrasts cannot be interpreted as independent effects or interactions. The Full configuration is retained for its declared guarded route and failure-case behavior rather than because every incremental stage lowers aggregate ATE.

GDOR also does not eliminate all local errors. Semantic remains competitive on several translational RPE rows, and GDOR is not uniformly best in rotational RPE. The positive claim is sequence-level ATE rescue and reduced failure under the declared protocols, not universal dominance for every local-motion metric.

C. Persistent Map Interpretation

DYN-19 directly audits the separation between recovered tracking support and persistent mapping weight. All 30 Full cells contain observed recovery and pass the zero-overlap route certificate, whereas all 30 Full-NoM counterfactual cells fail when recovered support is intentionally admitted to mapping. The 72-cell common-manifest evaluation adds a multi-seed output-level diagnostic and shows higher online static-region PSNR for Full on most paired cells. These results support the implemented route invariant and limited rendering preservation

TABLE III

SAFETY, EFFICIENCY, AND DYN-19 MAPPING DIAGNOSTICS. PANEL A REPORTS THE DYN-15 ALL7 MEAN. PANEL B REPRESENTS ALL 72 MAPPING CELLS: 12 SEQUENCE/SEED CELLS PER METHOD IN EACH POSE MODE. COMPLETENESS AND GHOST-RISK COLUMNS ARE OCCLUDED-BACKGROUND PROXIES, NOT INDEPENDENT GROUND TRUTH.

Panel	Method	ATE [cm] ↓	t-RPE [cm] ↓	Failure [%] ↓	Time [s/run] ↓	Map support [%]
A: DYN-15 All7	Semantic	11.331	4.277	0.785	32.46	100.00
	GDOR	3.798	3.945	0.144	49.84	63.31
	Strict	7.611	4.556	0.318	63.61	63.31
Panel B pose	Method	Static PSNR ↑	Static SSIM ↑	Proxy completeness ↑	Proxy ghost risk ↓	Cells
Online	Semantic	18.866	0.695	0.582	0.181	12
	MapMatched	18.990	0.700	0.579	0.184	12
	Full	19.763	0.728	0.582	0.188	12
GT-aligned	Semantic	12.082	0.492	0.402	0.351	12
	MapMatched	12.442	0.503	0.428	0.329	12
	Full	13.491	0.533	0.440	0.312	12

under the declared protocol. They do not constitute independent ghost-contamination ground truth, complete-background truth, or a universal mapping-superiority claim.

D. Reproducibility and External Validity

The clean source branch contains the GDOR implementation, configurations, tests, benchmark scripts, compact claim-bearing CSV files, and validators. The separate DYN-19 result release publishes the retained trajectories, masks, maps, renders, per-frame logs, certificates, summaries, plans, and checksums; datasets and model weights remain external dependencies. Every claim-bearing DYN-19 run is bound to the same clean source commit and frozen plan hash.

External validity and independent mapping truth remain the main blockers to a broader ranking. Published DyPhoSLAM numbers are stronger than the local GDOR values on the four TUM sequences, but the implementations and protocols are not matched. DL-SLAM also overlaps with the high-level tracking–mapping separation premise, so GDOR’s claim is restricted to its RGB-D/ORB routing mechanism and same-source evidence. The correct next evidence package is therefore a frozen eligible external-baseline rerun plus blind or external revealed-background annotation, not threshold adjustment on GDOR. Until those gates pass, published systems remain related-work context and proxy mapping metrics remain diagnostic.

VI. CONCLUSION

GDOR-SLAM implements an auditable two-route contract for dynamic-region evidence in an RGB-D/ORB Gaussian SLAM pipeline. Temporal depth, residual flow, and tracking-risk evidence qualify a conservative subset of excluded observations for transient localization, while persistent ORB and Gaussian mapping remain governed by the conservative map mask. This separation allows tracking utility without automatic map-writing authority.

The completed DYN-19 package extends evaluation to 90 main runs, 120 ordered-ablation runs, and 72 multi-seed mapping cells. Its results show conditional rescue of severe tracking failures rather than uniform improvement, and the non-monotonic ordered contrasts do not identify independent

component effects. The 30/30 Full certificates and the failing Full-NoM counterfactual support the implemented route invariant, while the mapping outputs remain diagnostic without independent ghost/completeness truth. The earlier single-seed DYN-16 audit is retained only as a limited preservation diagnostic.

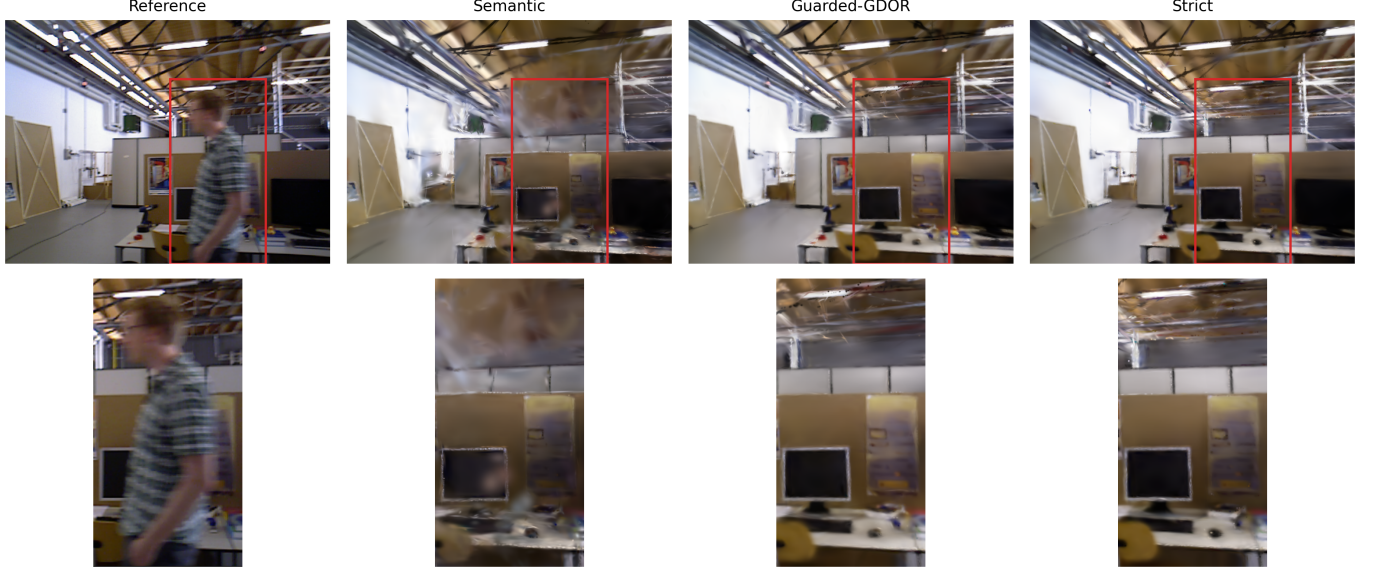
The current evidence supports the specific guarded recovery and routing mechanism; it does not claim conceptual priority for tracking–mapping separation, independent component causality, or universal dominance. It does not establish superiority over protocol-matched external implementations or measured zero ghost contamination. Eligible external reruns and independent revealed-background truth are the remaining requirements for broader claims.

REFERENCES

- [1] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, “3d gaussian splatting for real-time radiance field rendering,” *ACM Transactions on Graphics*, vol. 42, no. 4, 2023.
- [2] J. Huai, F. Xie, and H. Wang, “Photo-slam: Real-time simultaneous localization and photorealistic mapping for monocular, stereo, and rgb-d cameras,” in *CVPR*, 2024.
- [3] N. Keetha, J. Karhade, K. M. Jatavallabhula, G. Yang, S. Scherer, D. Ramanan, and J. Luiten, “Splatam: Splat, track & map 3d gaussians for dense rgb-d slam,” in *CVPR*, 2024.
- [4] C. Yan, D. Qu, D. Wang, D. Xu, Z. Wang, B. Zhao, and X. Li, “Gs-slam: Dense visual slam with 3d gaussian splatting,” in *arXiv preprint arXiv:2311.11700*, 2023.
- [5] V. Yugay, Y. Li, T. Gevers, and M. R. Oswald, “Gaussian-slam: Photo-realistic dense slam with gaussian splatting,” in *arXiv preprint arXiv:2312.10070*, 2023.
- [6] B. Bescos, J. M. Fácil, J. Civera, and J. Neira, “Dynaslam: Tracking, mapping, and inpainting in dynamic scenes,” *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 4076–4083, 2018.
- [7] C. Yu, Z. Liu, X.-J. Liu, F. Xie, Y. Yang, Q. Wei, and Q. Fei, “Ds-slam: A semantic visual slam towards dynamic environments,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018, pp. 1168–1174.
- [8] L. Xiao, J. Wang, X. Qiu, Z. Rong, and X. Zou, “Dynamic-slam: Semantic monocular visual localization and mapping based on deep learning in dynamic environment,” *Robotics and Autonomous Systems*, vol. 117, pp. 1–16, 2019.
- [9] Y. Xu, H. Jiang, Z. Xiao, J. Feng, and L. Zhang, “Dg-slam: Robust dynamic gaussian splatting slam with hybrid pose optimization,” in *Advances in Neural Information Processing Systems*, 2024. [Online]. Available: <https://arxiv.org/abs/2411.08373>
- [10] M. Kong, J. Lee, S. Lee, and E. Kim, “Dgs-slam: Gaussian splatting slam in dynamic environment,” *arXiv preprint arXiv:2411.10722*, 2024. [Online]. Available: <https://arxiv.org/abs/2411.10722>

TUM walking_xyz frame 202

Held-out frame 202 at 1341846320.631954908 s



Bonn crowd frame 201

Held-out frame 201 at 1548339826.611599922 s

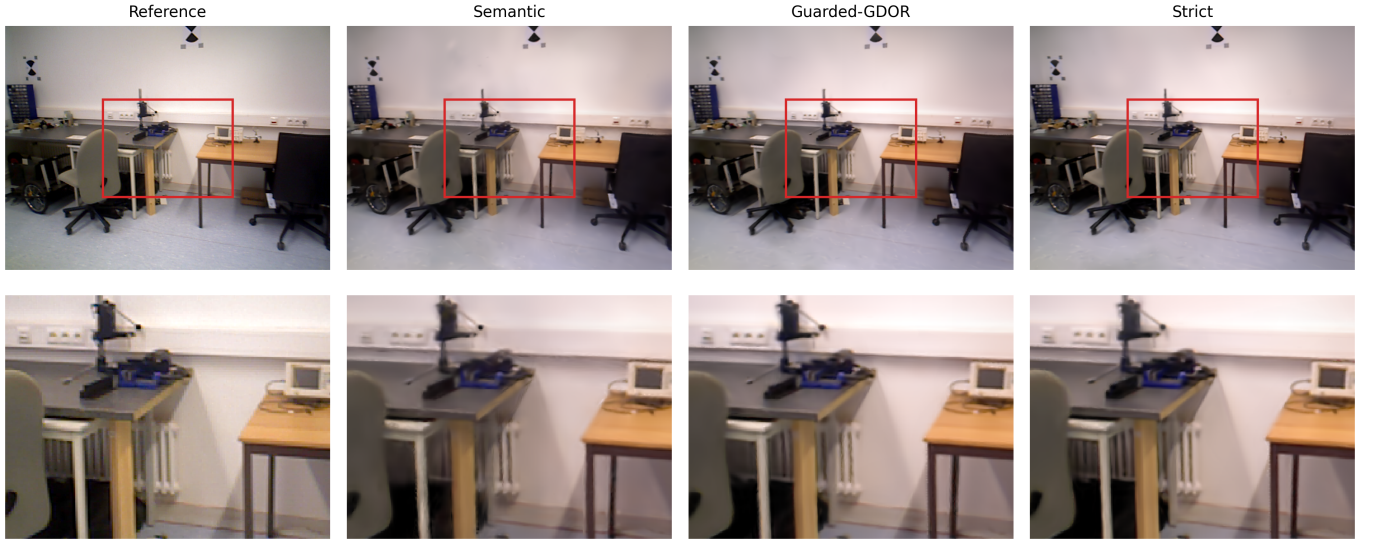


Fig. 5. DYN-16 common-view online mapping diagnostic. Each row uses the same held-out frame, fixed static-region mask, and final-map render for Semantic, GDOR, and Strict. This historical single-seed view is retained as a limited preservation diagnostic; the quantitative multi-seed mapping result is in Table III.

- [11] Y. Liu, K. Fan, B. Lan, and H. Liu, “Dypho-slam: Real-time photorealistic slam in dynamic environments,” in *IEEE International Conference on Multimedia and Expo*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.00741>
- [12] L. Zhang, Y.-A. Liu, X. Jiang, C. Huang, D. Li, and Y. Zhang, “Dags-slam: Dynamic-aware 3dgs slam via spatiotemporal motion probability and uncertainty-aware scheduling,” *arXiv preprint arXiv:2602.21644*, 2026. [Online]. Available: <https://arxiv.org/abs/2602.21644>
- [13] R. Mur-Artal, J. Montiel, and J. D. Tardós, “Orb-slam: A versatile and accurate monocular slam system,” *IEEE Transactions on Robotics*, vol. 31, no. 5, pp. 1147–1163, 2015.
- [14] C. Campos, R. Elvira, J. J. Rodríguez, J. M. Montiel, and J. D. Tardós, “Orb-slam3: An accurate open-source library for visual, visual-inertial, and multimap slam,” *IEEE Transactions on Robotics*, vol. 37, no. 6, pp. 1874–1890, 2021.
- [15] T. Whelan, R. F. Salas-Moreno, B. Glocker, A. J. Davison, and S. Leutenegger, “Elasticfusion: Real-time dense slam and light source estimation,” *International Journal of Robotics Research*, vol. 35, no. 14, pp. 1697–1716, 2016.
- [16] W. Tan, H. Liu, Z. Dong, G. Zhang, and H. Bao, “Robust and accurate slam in unstructured environments,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2013, pp. 5459–5466.
- [17] E. Palazzolo, J. Behley, P. Lottes, P. Giguere, and C. Stachniss, “Refusion: 3d reconstruction in dynamic environments for rgb-d cameras exploiting residuals,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019, pp. 3193–3200.
- [18] J. Zhang, M. Henein, R. Mahony, and V. Ila, “Vdo-slam: A visual dynamic object-aware slam system,” in *arXiv preprint arXiv:2005.11052*, 2020.
- [19] B. D. Lucas and T. Kanade, “An iterative image registration technique with an application to stereo vision,” in *International Joint Conference on Artificial Intelligence (IJCAI)*, 1981, pp. 674–679.
- [20] E. Sucar, S. Liu, J. Ortiz, and A. J. Davison, “imap: Implicit mapping and positioning in real-time,” in *ICCV*, 2021, pp. 6229–6238.
- [21] Z. Zhu, S. Peng, V. Larsson, W. Xu, H. Bao, Z. Cui, M. R. Oswald, and M. Pollefeys, “Nice-slam: Neural implicit scalable encoding for slam,”

- in *CVPR*, 2022, pp. 12 786–12 796.
- [22] X. Yang, H. Li, H. Zhai, Y. Ming, Y. Liu, and G. Zhang, “Vox-fusion: Dense tracking and mapping with voxel-based neural implicit representation,” in *IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, 2022, pp. 499–507.
 - [23] X. Hu, C. Zhang, M. Zhao, Y. Gui, X. Zhang, and X. Ji, “Dygs-slam: Real-time accurate localization and gaussian reconstruction for dynamic scenes,” in *International Conference on Computer Vision*, 2025, pp. 9561–9571. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2025/html/Hu_DyGS-SLAM_Real-Time_Accurate_Localization_and_Gaussian_Reconstruction_for_Dynamic_Scenes_ICCV_2025_paper.html
 - [24] C. Huang, L. Zhang, T. Deng, K. Wang, and M. Li, “Motion probability-based 3dgs-slam in dynamic environment,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025, pp. 14 045–14 052. [Online]. Available: <https://ieeexplore.ieee.org/document/11247059>
 - [25] Z. Xu, Q. Li, X. Liu, C. Chen, and J. Niu, “DI-slam: Enabling high-fidelity gaussian splatting slam in dynamic environments based on dual-level probability,” *arXiv preprint arXiv:2607.01860*, 2026. [Online]. Available: <https://arxiv.org/abs/2607.01860>
 - [26] S. Zhu, Y. Huang, W. Wu, C. Jiang, Y. Chen, I.-M. Chen, and H. Wang, “D²gslam: 4d dynamic gaussian splatting slam,” *arXiv preprint arXiv:2512.09411*, 2025. [Online]. Available: <https://arxiv.org/abs/2512.09411>
 - [27] J. Sturm, N. Engelhard, F. Endres, W. Burgard, and D. Cremers, “A benchmark for the evaluation of rgb-d slam systems,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2012, pp. 573–580.